

Software Requirements Specification

for

**Electric Vehicle Charging Station
Management System**

Release 1.0

Version 1.0 approved

Prepared by Group 4

Process Impact

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Table of Content

1. Introduction	2
1.1 Purpose	2
1.2 Document Conventions	2
1.3 Project Scope	2
1.4 References	2
2. Overall Description	3
2.1 Product Perspective	3
2.2 User Classes and Characteristics	3
2.3 Operating Environment	4
2.4 Design and Implementation Constraints.....	5
2.5 Assumption and Dependencies	5
3. System Features	6
3.1 Use Case Diagram	6
3.2 Use Case Specification	7
4. Data Requirements	17
4.1 Logical Data Model	17
4.1.1 Conceptual Model.....	17
4.1.2 Logical Diagram	18
4.2 Data Dictionary	19
4.3 Business Rule	25
5. External Interface Requirements	27
5.1 User Interfaces	27
5.2 Software Interface	28
5.3 Hardware Interface	28
5.4 Communications Interface	29
6. Quality Attributes	29
6.1 Usability.....	29
6.2 Performance	29
6.3 Security	30
6.4 Safety	30
6.5 Availability Requirements	30
6.6 Robustness Requirements	30

1. Introduction

1.1 Purpose

This SRS document specifies the requirements for EV Charging Station Management System, version 1.0, for use by development team, QA, PM, station operations, business department and technical writer for design, implementation, testing, acceptance and maintenance

1.2 Document Conventions

This document follows a set of conventions to ensure clarity, consistency, and readability across all sections of the Software Requirements Specification (SRS) for the *EV Charging Station Management System (EVCSMS)*.

Dates follow format MM/DD/YYYY, currency defaults to VND, times are 24-hour; UI strings may appear in English side by side for clarity.

Actors are EV Driver, CS Staff, and Admin; nonfunctional items carry tags like SEC, PERF, and UX for traceability.

1.3 Project Scope

The system provides functions for discovering, reserving, controlling charging sessions, making payments, issuing invoices, monitoring, and managing EV charging stations through a web platform. Its goal is to help users and operators improve utilization efficiency, revenue, and user experience across multiple locations.

Key features include: a charging station map with search filters, reservation and session start via QR code, real-time charging status and notifications, payment by cash or e-wallet, electronic invoicing, onsite staff support tools, remote control, role-based administration, and reporting and analytics.

1.4 References

1. Electric Vehicle Charging Station Management System Vision and scope, [EVCSMS Vision and scope - Google Tài liệu](#)
2. VNPay. *VNPay Payment Gateway Documentation*. Retrieved November 10, 2025, <https://vnpay.vn>
3. Leaflet. *Leaflet - JavaScript Library for Interactive Maps*. Retrieved November 10, 2025, from: <https://leafletjs.com>

2. Overall Description

2.1 Product Perspective

This is a new system integrating with OCPP chargers, payment gateways, and mapping, while exposing partner APIs for fleets and third parties within a broader e-mobility ecosystem.

It can stand alone or serve as the control component of a larger energy or mobility platform, with interfaces to external identity and billing when needed.

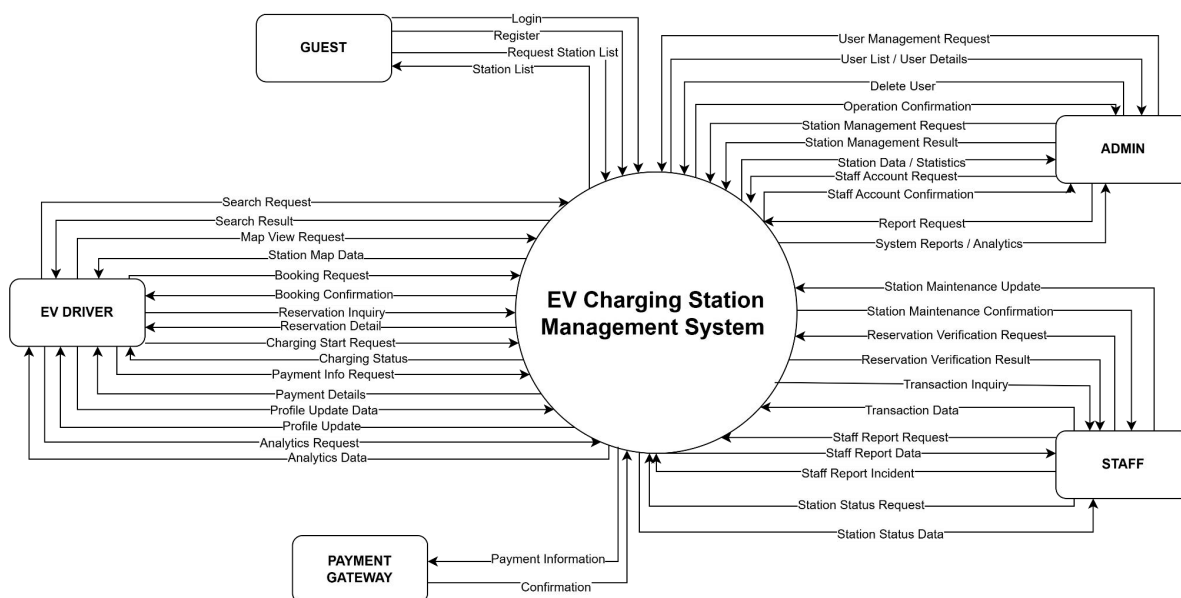


Figure 1. Context diagram for EV Charging Station Management System

2.2 User Classes and Characteristics

EV Driver	An EV Driver is the primary user of the charging-station ecosystem who locates charging stations on the in-web map, books and manages charging sessions, starts and terminates sessions via Web/QR, tracks charge progress and receives notifications, and completes payments via Vnpay platform. There are several hundred to thousands of potential EV Owners as the system scales; regular users may charge multiple times per week with usage spikes at commuting hours and before long trips. EV Drivers typically require no formal training (in-app guidance only), need reliable low-latency map and session flows (e.g.,
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	booking success $\geq 98\%$, map load $< 2s$).
Station Staff	Station Staff are on-site operators responsible for daily station operations including monitoring station health and alarms, assisting users with session start/stop or manual overrides, performing basic hardware resets and maintenance tasks, and updating local logs through a staff dashboard. Staff interact with the system daily and use a staff web dashboard most will require onboarding and periodic training for hardware/software procedures. Their role needs offline-capable workflows and fast alarm notification latency (real-time or near real-time).
Administrator	System Administrators configure backend systems, manage user permissions and security policies, monitor system health dashboards and run analytics and predictive planning (demand forecasts, station utilization). They typically operate during business hours with on-call responsibilities for outages, require elevated access to system consoles and logs, and depend on reliable telemetry and data integrity for model training and validation. They also require access controls, audit trails for critical operations, and tools for anomaly detection and incident response.

2.3 Operating Environment

OE-1: The EV CSMS shall operate correctly with the following web browsers on PC/laptop: Google Chrome (latest versions), Apple Safari (latest versions), Microsoft Edge (latest versions) and CocCoc. All user interactions are through a web console.

OE-2: The CSMS backend shall operate on servers running corporate- approved Linux distributions (e.g., Red Hat Enterprise Linux) with a supported web server/reverse proxy (e.g., Apache HTTP Server or Nginx) and Java/Spring Boot runtime per platform standards.

OE-5: The system shall provide secure access over the corporate Intranet and via VPN from the public Internet for Station Staff and Administrators. No mobile apps are required.

OE-4: Charge points shall connect to CSMS over secure IP networks (wired or cellular) using OCPP 1.6J, with time synchronization via NTP and TLS where supported.

OE-5: Databases shall run on the corporate- approved SQL server version with high availability options enabled per infrastructure policy.

2.4 Design and Implementation Constraints

CO-1: System design, code, and maintenance documentation shall conform to the corporate Software Development Standards and secure coding guidelines.

CO-2: The CSMS shall use the corporate- standard relational database engine (SQL Server) and approved messaging/cache services.

CO-3: The CSMS shall implement OCPP 1.6J for charge point communications in v1.0, with an upgrade path to OCPP 2.0.1.

CO-4: Payment integrations shall comply with PCI DSS; no cardholder data shall be stored on CSMS components.

CO-5: E- invoices shall comply with Vietnamese e- invoicing regulations and use organization- approved digital certificates.

CO-6: Role- based access control (RBAC) and audit logging shall be implemented across Admin, Station Staff, and Driver roles.

2.5 Assumptions and Dependencies

AS-1: The system assumes that all EV charging stations have a stable Internet connection to send and receive real-time charging data.

DE-1: The operation of the system depends on the network infrastructure of charging stations to ensure continuous and accurate data synchronization.

DE-2: The operation of the system depends on the charging hardware's API and control software to exchange and process charging data correctly.

3. System Features

3.1 Use Case diagram

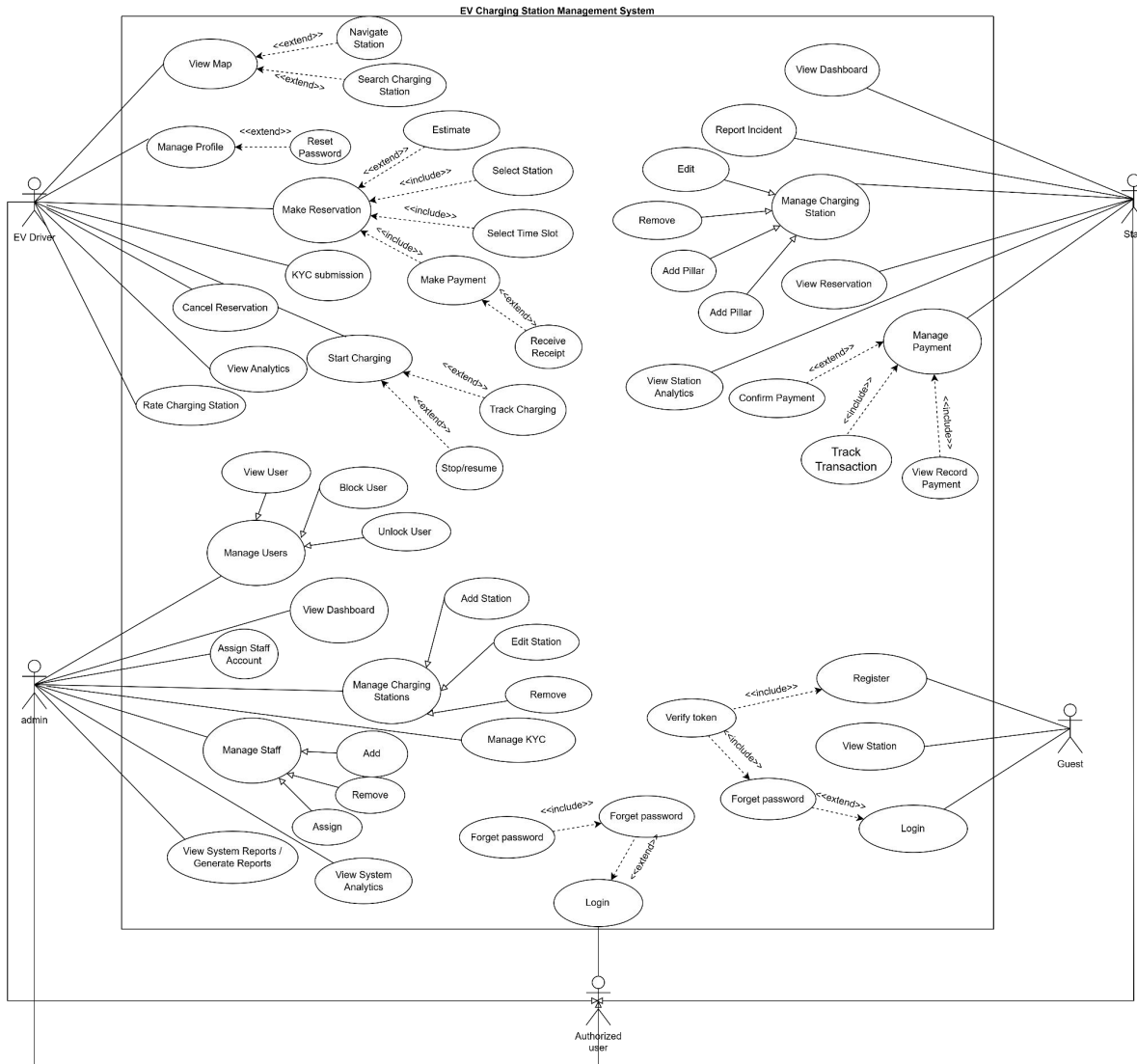


Figure 2. Use case diagram for EV Charging Station Management System

3.2 Use Case Specification

ID and Name:	UC-1 Booking Station		
Created By:	Hoang	Date Created:	26/9/25
Primary Actor:	Driver	Secondary Actors:	Booking System
Description:	A driver opens the EV Charging Management application to find the nearest available charging station, reserves a charging point, and scans a QR code on-site to start the charging session. The system displays the battery status (SOC %), remaining time, and cost, and sends a notification when the battery is fully charged. The driver then pays through an e-wallet or other available payment methods.		
Trigger:	The driver wants to charge the EV and opens the application to make a reservation.		
Preconditions:	PRE-1. The driver is registered and logged into the system. PRE-2. The driver has a valid payment method (e-wallet, bank account, or subscription package).		
Postconditions:	POST-1. The charging session is recorded in the system with status "In Progress" or "Completed." POST-2. The system updates the charging point status (busy/available). POST-3. The system records the payment transaction and updates the driver's history.		

ID and Name:	UC-1 Booking Station
Normal Flow:	1.0 Booking Station 1. The driver opens the application and views the charging station map. 2. The system displays a list of stations by location, power, availability, and pricing. 3. The driver selects an available charging point. 4. The driver makes a reservation. 5. The system confirms the reservation and holds the charging point for a limited period. 6. The driver arrives at the station and scans the QR code to start charging. 7. The system displays SOC (%), power, estimated cost, and remaining time. 8. When the battery is full or the driver manually stops, the system ends the charging session. 9. The system calculates the cost and issues an electronic invoice. 10. The driver pays via e-wallet / online banking. 11. The system stores the charging session and updates the charging point status to "available."
Alternative Flows:	1.1.1 Book multiple sessions in advance The driver requests to book another charging session for another date. Return to step 1 of the normal flow.
Exceptions:	1.0.E1 No available station near requested location The system informs: "No available charging points in this area." 2a. If the driver cancels → the system terminates the use case. 2b. If the driver searches for another station → return to step 2 of the normal flow. 1.0.E2 Charging point malfunction (offline / not responding) The system informs: "This charging point is currently unavailable." 2a. If the driver cancels → the system terminates the use case. 2b. If the driver selects another point within the same station →

ID and Name:	UC-1 Booking Station
	return to step 3 of the normal flow.
Priority:	High
Frequency of Use:	Daily, many drivers use it. Peak hours between 5:00 PM – 9:00 PM.
Business Rules:	BR-1: The driver can only hold a reservation for a limited time (e.g., 15 minutes). BR-2: Payment must be completed before the system records the charging session as “Completed.” BR-3: If the driver does not arrive within the reservation period, the charging point will be released back to the system. BR-4: A driver cannot book overlapping charging sessions. For example, if a reservation is already made for 12:00, the next one must be at 12:30 or later.
Other Information:	1. The driver can view charging history and costs for the last 6 months. 2. The system sends notifications when the battery is full or nearly full (depending on settings).
Assumptions:	Assume 70% of drivers choose online payment, 30% pay at the station. Charging infrastructure can synchronize status in real time.

ID and Name:	UC-2 Payment
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Created By:	Pham	Date Created:	29/06/25
Primary Actor:	EV Driver	Secondary Actors:	Payment Gateway
Description:	The EV Driver requests to charge at a station/connector. The system displays an estimated cost (based on kWh, time, or session). The driver selects a payment method (prepaid wallet, card, QR/PayNow, postpaid, membership). The system performs authorization/capture, either allowing the charging session to start or rejecting it, records the transaction, and sends an invoice/receipt.		
Trigger:	The EV Driver initiates a charging session that requires payment authorization.		
Preconditions:	PRE-1. The driver has logged in or has been authenticated. PRE-2. The driver has a valid payment method. PRE-3. The charging connector is in Available/Reserved status for the driver. PRE-4. The payment gateway is operational. PRE-5. The charging station supports billing communication and is online.		
Postconditions:	POST-1. The payment transaction is recorded with a status (Authorized / Captured / Failed). POST-2. The connector status is updated (Charging / Occupied / Available). POST-3. An invoice/receipt is sent (via email / app notification). POST-4. Wallet balance (if used) and subscription quota are updated.		

Normal Flow:	Charging Session 1. The driver selects a station/connector to start charging. 2. The system displays the estimated cost: estimated kWh, unit price, fixed fees, taxes, and total estimate. 3. The driver selects a payment method. 4. The system sends an authorization request to the Payment Gateway (or checks wallet balance, depending on the method). If authorization fails → Payment Failed. 5. If authorization succeeds, the system reserves the connector for the driver (locks the session) and allows charging to begin. 6. While charging, the system records statistics (start_time, energy_delivered in kWh, power profile). 7. When the session ends (driver stops or auto-stop), the system calculates the actual price: final_kWh * unit_price + fees + taxes. 8. The system charges the final amount (or creates an invoice for postpaid). 9. The system records the transaction with details (transaction_id, amounts, method, provider_ref). 10. The system sends a receipt/invoice to the driver and updates the connector status page.
Alternative Flows:	1. Prepaid Wallet Flow The user preloads funds into a wallet. At session start, the system checks the balance; if sufficient, it deducts/holds the estimated amount. After the session, it deducts the actual cost and refunds any excess. 2. Membership / Subscription Flow If the driver has a subscription package, the system applies benefits (discounts, free kWh within limits). Only over-limit usage is charged. Periodic subscription fees are managed separately. 3. Postpaid Flow For enterprise accounts: sessions are tracked but not charged immediately. Invoices are generated periodically (weekly/monthly) and sent to the billing system. Requires credit limit & KYC.
Exceptions:	E1. Authorization Failure: Display error, request alternative payment method, cancel reservation. E2. Capture Failure after Session: Send unpaid invoice, temporarily lock account if over limit, notify driver and support operations.
Priority:	High

Business Rules:	BR1: Valid Payment Method The EV Driver must have at least one valid payment method registered. BR2: Pre-Authorization The system must perform authorization (or wallet balance check) for the estimated amount before the charging session starts. BR3: Membership/Subscription Application If the driver has a membership or subscription, the system must apply benefits before calculating the final charge. BR4: Invoice/Receipt Delivery An invoice or receipt must be delivered immediately after payment
Other Information:	1. Users can view charging history and invoices for the past 6 months and reuse a previous session as a template (e.g., same station, similar duration) if available. 2. The system sends receipts/invoices immediately after capture; for postpaid accounts, invoices are consolidated per cycle (e.g., monthly) and include session details.
Assumptions	The payment gateway and charging station remain available and connected throughout the transaction, ensuring that authorization, charging, and capture processes can be completed without interruption

ID and Name:	UC-3 Reserve & Start Charging Session		
Created By:	Huy	Date Created:	26/9/2025
Primary Actor:	EV Driver		
Description:	The system provides a map displaying charging stations based on location, power capacity, availability status (available/in use), connector type, charging speed, and price. The driver can select a suitable station, reserve a charging point, go to the station, and use a QR code scan or app authentication to start the charging session. During the session, the application displays detailed status (SOC %, remaining time, cost) and sends notifications when the battery is fully charged.		

Trigger:	The EV driver can reserve and start a charging session at a station.
Preconditions:	PRE-1. The driver is logged into their account. PRE-2. The driver has verified their complete profile information. PRE-3. The vehicle's GPS functionality is working properly.
Postconditions:	POST-1. The charging point is successfully reserved and held for a specified period of time. POST-2. The charging session is started at the selected station. POST-3. Charging status, remaining time, and cost are fully displayed.
Normal Flow:	1.The EV Driver opens the application and enables GPS. 2.The system displays a map with nearby charging stations, including details (capacity, status, connector type, charging speed, price). 3.The EV Driver selects a charging station and requests a reservation. 4.The system checks availability and confirms the reservation (e.g., for 15 minutes). 5.Upon arrival at the station, the EV Driver scans a QR code or authenticates via the app to start the charging session. 6.The system starts the charging session and displays status information: SOC %, remaining time, cost. 7.When the battery is full or the EV Driver chooses to stop charging, the system ends the session, calculates the cost, and sends a notification.
Alternative Flows:	3a. No charging connectors available → The system notifies the driver and suggests alternative nearby stations. 5a. QR code error or connection failure → The EV Driver retries or selects another authentication method. 6a. Charging session encounters an issue (power outage, station malfunction) → The system stops charging, notifies the driver, and processes refunds if necessary.

Exceptions:	E1. No charging stations found within the allowed radius → display an error message. E2. GPS/Internet unavailable → display a message and request a retry. E3. Reservation time expired → the spot is released, and the EV Driver must reserve again.		
Priority:	High		
Frequency of Use:	Around 300 users, on average once per day. Peak load for this use case occurs between 9:00 AM and 10:00 AM local time.		
Business Rules:	BR-1: The system must always sort the station list by closest distance from the EV Driver's location. BR-2: Reservations are only possible if at least one charging connector is available. BR-3: Default reservation hold time is 15 minutes. BR-4: Charging status (SOC %, remaining time, cost) must be updated in real time.		
Other Information:	1.The EV Driver may cancel a reservation anytime before charging begins. 2.The EV Driver receives push notifications when the battery is nearly full or fully charged. 3.The system may integrate Google Maps or OpenStreetMap to assist with navigation to the station.		
Assumptions:	It is assumed that most charging stations support QR code scanning and app-based authentication. It is assumed that GPS and Internet connectivity are sufficiently stable to provide real-time data.		

ID and Name:	UC-4 Monitoring and Reporting Station Status		
Created By:	Bảo	Date Created:	26/9/25

Primary Actor:	EV-Admin	Secondary Actors:	EV-Staff
Description:	Admin and staff can monitor the online/offline status of each device in the charging station. The system continuously collects device status data, displays it on a real-time dashboard, and sends alerts when malfunctions, signal loss, or hardware errors occur.		
Trigger:	A device status changes (offline/online, malfunction, or signal loss).		
Preconditions:	PRE-1. The charging station has been configured in the system. PRE-2. Monitoring system and APIs are available. PRE-3. The user has Admin or Staff privileges.		
Postconditions:	POST-1. Device status is fully recorded in the system. POST-2. Malfunction alerts are delivered to Admin/Staff in a timely manner.		
Normal Flow:	1. The system continuously collects device status from the station. 2. Admin/Staff logs into the dashboard. 3. The dashboard displays online/offline status of devices. 4. When a malfunction occurs (offline, signal loss, hardware failure), the system generates a real-time alert. 5. Admin/Staff receives the alert on the dashboard or through the notification service. 6. Admin/Staff acknowledges and proceeds with troubleshooting. 7. Once the issue is resolved, the dashboard updates the device status back to normal.		
Alternative Flows:	Temporary network disconnection 1.1 If monitoring connection is interrupted, the system stores temporary data and resynchronizes once the connection is restored. Device sends data in incorrect format 2.1 The system discards the faulty data and notifies Admin/Staff to check the device. Notification service malfunction 3.1 The system still logs the alert, but the alert only appears on the dashboard and is not sent externally.		

Exceptions:	E1. Device not responding E2. Device fails to respond within the allowed time → system marks device as Offline. E3. Monitoring API unavailable E4. The system issues an alert “Monitoring system down”. E5. Error message is displayed, and user is asked to reload. E6. GPS or location data missing E7. Duplicate device ID detected. Monitoring service rejects duplicated input and notifies Admin to verify device configuration. E8. Notification delivery failure. If push notification fails (e.g., mobile app offline), the system retries delivery and logs failure events.
Priority	High
Frequency of Use	Continuous (24/7). Peak usage during heavy station load (9:00 AM – 12:00 PM).
Business Rules	1. Every device must be monitored for online/offline status. 2. Alerts must be delivered within <2s from detection. 3. Alert accuracy rate ≥ 95%.
Other Information:	1. Pain point: Management is unaware of real station/device status. 2. Checkpoints: anomaly detection, notification retry. 3. Deployment: real-time logging. 4. Suggestions: AI for predictive maintenance and early fault detection.
Assumptions	Devices at the charging station support sending status data continuously and in the correct format.

4. Data Requirements

4.1 Logical Data Model

4.1.1 Conceptual model

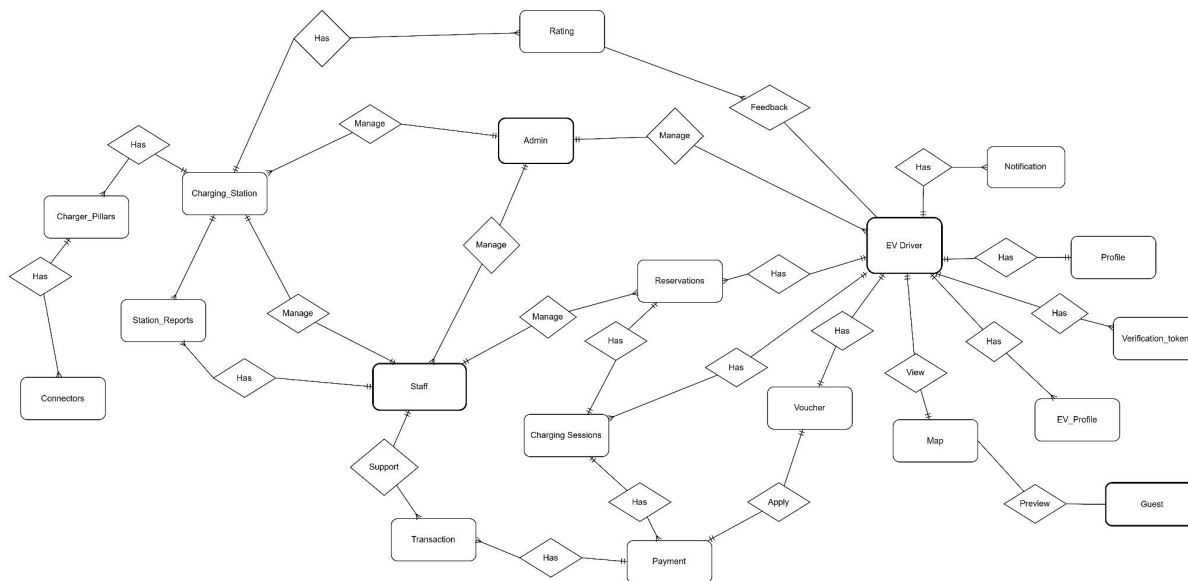


Figure 3. Conceptual model for EV Charging Station Management System

4.1.2 Logical Diagram

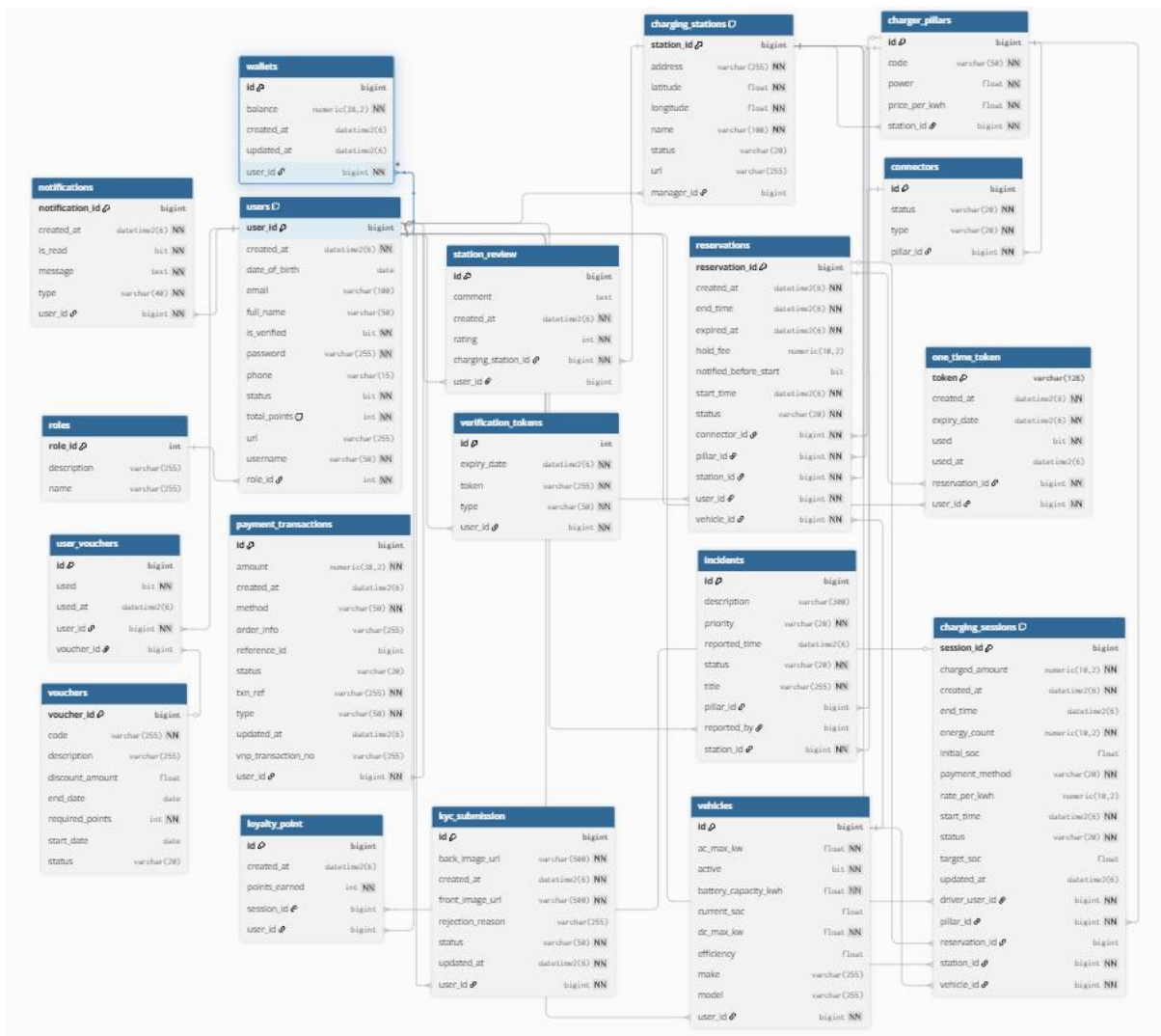


Figure 4. Logical diagram for EV Charging Station Management System

4.2 Data Dictionary

User entity

Data Element	Description	Composition or Data Type	Length	Values
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user_id	Unique identifier for each user in the system	integer		Primary key; auto-increment; unique
username	Username for login	varchar	50	Alphabetic characters only; not null
full_name	Full name of the user	varchar	50	Alphabetic characters only; not null
phone	Phone number of user	varchar	15	Numeric only; unique
email	User's registered email address used for login and notifications	varchar	100	Must be unique; must contain "@"; validated by regex
password	Hashed password string stored securely	varchar	255	Hashed using bcrypt or equivalent; not null
status	Indicates if the user account is active	boolean		Default = true; not null
created_at	Timestamp indicating when the user account was created	timestamp		Auto-generated by system on registration
role_id	Reference ID linking to user role definition (Driver, Staff, Admin)	integer (FK)		Foreign key -> roles.role_id; required
ev_id	Identifier of the EV linked to the user account	integer		Foreign key → evs.ev_id; nullable if user has no EV
date_of_birth	Date of birth of the user	date		Format: MM/DD/YYYY;

				optional field
is_verified	Indicates whether the user has completed email/OTP verification	boolean		TRUE = verified; FALSE = unverified (default)
url	URL to user profile picture	varchar		Optional; valid URL format
total_points	Loyalty or reward points of the user	int		Default = 0; not null
managed_station_id	Charging station managed by the user (if any)	integer (FK)		
charging_session_id	Associated charging sessions of the user	integer (FK)		Optional; references ChargingSession table; one-to-many relationship

Data Element	Description	Composition or Data Type	Length	Values
role_id	Unique identifier for each role in the system	integer		Primary key; auto-increment; unique
name	Name of the role	varchar	100	Alphabetic characters only; not null
description	Description of the role and its permissions	varchar	255	Optional; text describing the role
users	List of users associated with this role	list<User>		One-to-many relationship with User entity

Data Element	Description	Composition or Data Type	Length	Values
station_id	Unique identifier for each charging station in the system	integer		Primary key; auto-increment; unique
station_name	Official name or label of the charging station as displayed in the app	varchar	100	Required; may include letters, digits, and spaces
address	Full physical address of the charging station (street, city, province)	varchar	255	Required; commas and special characters allowed
latitude	Geographic latitude coordinate of the station	decimal(9,6)	9 digits total, 6 after decimal	Range: - 90.000000 to +90.000000
longitude	Geographic longitude coordinate of the station	decimal(9,6)	9 digits total, 6 after decimal	Range: - 180.000000 to +180.000000
status	Current operational state of the station	varchar	20	Possible values: Active, Offline, Maintenance, Decommissioned

Data Element	Description	Composition or Data Type	Length	Values
payment_id	Unique identifier for each payment in the system	integer		Primary key; auto-increment; unique

invoice_id	Unique identifier for each invoice in the system	integer		Foreign key → invoices.invoice_id ; nullable if payment not invoiced yet
method	Payment method selected by user or staff	varchar	20	Cash, card, ewallet, bank_transfer, voucher, mixed
provider	Payment provider or acquiring channel	varchar	30	VNPay sandbox
amount	Total amount charged for the payment (tax included)	decimal(5,2)	5 digits, 2 decimals	currency default VND
status	Lifecycle status of the payment	varchar	20	pending, authorized, captured, settled, failed, canceled, refunded, partially_refunded
transaction_ref	External gateway/acquirer transaction reference	varchar	80	Unique per provider; nullable for cash
created_at	Payment creation timestamp (UTC)	timestamp		NOT NULL; default current_timestamp

Data element	Description	Data type	Length	Values
ev_id	Unique identifier for each electric vehicle	BIGINT IDENTITY(1,1)	16 bytes	Primary key; auto-increment; unique
brand	The manufacturer or brand name of the electric vehicle	NVARCHAR(255)	Up to 255 chars	
model	The model name of the electric vehicle	NVARCHAR(255)	Up to 255 chars	
batteryCapacityKwh	Total battery capacity in kilowatt-hours (kWh)	DECIMAL(5,2)	5 digits, 2 decimals	0.1 - 100
acMaxKw	Maximum AC charging power (in kilowatts)	DECIMAL(5,2)	5 digits, 2 decimals	0.1 - 22
dcMaxKw	Maximum DC charging power (in kilowatts)	DECIMAL(5,2)	5 digits, 2 decimals	0.1 - 350
efficiency	Energy efficiency (km/kWh or mi/kWh)	DECIMAL(4,2)	4 digits, 2 decimals	3.0 - 7.0 km/kWh

Data Element	Description	Composition or Data Type	Length	Values
plan_id	Unique identifier for each subscription plan	integer		Primary key; auto-increment; unique

name	Name of the subscription plan as displayed to users	varchar	50	Required; letters, digits, and spaces allowed
price	Price of the plan in the system's default currency	decimal(10,2)		Must be a positive value (e.g., 19.99, 59.00)
billing_cycle	Frequency of billing for the plan	varchar	10	Possible values: Monthly, Quarterly, Yearly
included_kwh	Amount of energy (in kWh) included in the plan	decimal(10,2)		Must be ≥ 0 ; represents energy quota
description	Additional information or details about the plan	text		Optional; may include HTML formatting or line breaks
created_at	Timestamp of when the plan was created	timestamp		Auto-generated on record creation

4.3 Business Rule

ID	Rule Definition
BR-01	The system must allow users to register using email, phone number, or social login (Google).
BR-02	The system must verify the user's email or phone number via OTP or verification link before account activation.
BR-03	The system must support user login via username/password and OAuth-based social login.
BR-04	The system must issue access and refresh tokens (JWT) to maintain user sessions.
BR-05	When an access token expires, the system must return HTTP 401; if a valid refresh token is provided, a new access token must be issued.
BR-06	The system must implement Role-Based Access Control (RBAC) with roles: EV_DRIVER, CS_STAFF, and ADMIN
BR-07	Users without sufficient permissions must receive HTTP 403 when attempting restricted actions.
BR-08	The system must allow users to reset passwords through email or OTP verification.
BR-09	The system must limit OTP attempts and throttle password reset requests to prevent brute-force attacks.
BR-10	Users must be able to update personal profiles and vehicle information.
BR-11	Functions requiring verified identity (e.g., corporate invoices) must be locked until KYC is completed.
BR-12	The system must update station and charger status in real-time when heartbeat signals are received.
BR-13	The system must store connector details, including type (CCS, CHAdeMO, AC), power rating (kW), number of ports, and pricing.
BR-14	APIs must return accurate connector types and power data, with filtering options by type and power.
BR-15	Authorized users (ADMIN, CS_STAFF) must be able to remotely control chargers (turn on/off, open/close ports).
BR-16	Each remote control command must generate an audit log entry recording who performed the action and when.

BR-17	Offline stations must reject new charging sessions while allowing existing sessions to reconnect or report errors.
BR-18	Users must be notified when stations go offline or reconnect.
BR-19	The system must allow users to search for charging stations by location radius and apply filters (connector type, price, status, availability).
BR-20	Search results must be sorted by distance and availability by default.
BR-21	Each search result must display key information: distance, available ports, power, price, and last status update.
BR-22	The system must display “last updated at X” to indicate data freshness for users.
BR-23	The system must allow drivers to reserve a charging port for a limited period (e.g., 15 minutes).
BR-24	Booking statuses must include PENDING, CONFIRMED, CANCELLED, and EXPIRED.
BR-25	If a user does not arrive within the hold period, the booking status must automatically change to EXPIRED.
BR-26	The system must apply locking mechanisms to prevent double-booking (e.g., optimistic/pessimistic locking).
BR-27	When two users attempt simultaneous bookings for the same port, only one booking must succeed.
BR-28	A charging session must only start after payment authorization (if required) and charger confirmation.
BR-29	Charging sessions can start via QR code scan or remote command through the app.
BR-30	If payment authorization fails, the session must not start.
BR-31	Users must be able to pause, resume, or end charging sessions depending on station capability.
BR-32	The system must record session data including start/end time, kWh consumed, duration, and estimated cost.
BR-33	Users must be able to cancel bookings; late cancellations may incur fees per policy.
BR-34	No-show or late-cancel penalties must be automatically applied according to configuration (e.g., deposit retention).
BR-35	If a charger disconnects unexpectedly, the system must mark the session as

	INTERRUPTED and attempt automatic reconnection.
BR-36	All interruption and retry events must be logged, and users or CS_STAFF must be notified.
BR-37	The system must support multiple pricing models: per kWh, per minute, or fixed slot plus extra parking fees.
BR-38	Pricing must be configurable by station, time period, and peak hours.
BR-39	Invoices must correctly apply the selected pricing formula for each session.
BR-40	Before a session starts, the system must create a prepayment or hold deposit on the payment service provider (PSP).
BR-41	At session end, the system must capture the actual amount, refunding any excess automatically.
BR-42	Supported payment methods must include e-wallets, credit/debit cards, bank transfer, and QR code payment.
BR-43	Each payment transaction must record its PSP transaction ID and status (PENDING, SUCCESS, FAILED).
BR-44	Sessions cannot start if the payment status is FAILED or mismatched.
BR-45	Refunds must be supported for device errors or incorrect charges.
BR-46	Refund and dispute workflows must be traceable and resolved within the defined SLA (e.g., 5 days).
BR-47	After successful payment, users must be able to request an electronic VAT invoice.
BR-48	All invoices must include billing details and be stored for future reference.
BR-49	The system must automatically change a reservation from PENDING to EXPIRED if its createdAt timestamp is more than 5 minutes in the past and no other state transition has occurred.
BR-50	When a reservation transitions to EXPIRED for any reason, the system must automatically release the associated connector by setting its status back to AVAILABLE, provided there is no active charging session bound to that connector.
BR-51	When the current time reaches or passes a reservation's configured startTime, the system must transition that reservation into VERIFYING state (unless it is already CANCELLED or EXPIRED) to indicate that the check-in/plug-in window has opened.
BR-52	If the current time is more than 15 minutes after a reservation's startTime and the reservation is still in any intermediate check-in state (e.g., VERIFYING, VERIFIED, or PLUGGED), the system must automatically change the reservation

	status to EXPIRED to enforce the late-arrival policy.
BR-53	When the current time passes a reservation's configured endTime, any reservation that is not already COMPLETED, CANCELLED, or EXPIRED must be automatically transitioned to EXPIRED
BR-54	The system must send a reminder notification to the reservation's driver user approximately 5 minutes before the reservation startTime, informing them of the upcoming booking and expected arrival time.

5. External Interface Requirements

5.1 User Interfaces

UI-1: The EV Charging Station Management System user interface shall be designed for internal web use, ensuring consistent layout and user experience across all pages.

UI-2: The system shall include a navigation bar allowing quick access to main sections such as Charging Stations, Transactions, User Management, and Dashboard.

UI-3: The system shall display charger pillar statuses — Available, Charging, Reserved, Offline — using unified colors and icons across the platform.

UI-4: The user interface shall display account information (name, role, and avatar) at the top of the page to indicate the current login status.

UI-5: The booking confirmation screen shall display a countdown timer (e.g., 15 minutes) showing how long the reservation will be held before automatic release.

UI-6: The system shall display a map view of charging station locations using latitude and longitude coordinates to help users find nearby stations.

5.2 Giao diện phần mềm (Software Interfaces)

SI-1: Cổng thanh toán (Payment Gateway)

SI-1.1: Hệ thống EVCSMS phải giao tiếp với cổng thanh toán VNPay thông qua RESTful API qua HTTPS để thực hiện các thao tác xác thực, thanh toán, và hoàn tiền.

SI-1.2: VNPay phải trả về mã giao dịch (vnp_TxnRef) và mã phản hồi (vnp_HashCode), và hệ thống phải lưu trữ các thông tin này cùng với bản ghi thanh toán.

SI-1.3: Hệ thống phải xử lý callback bất đồng bộ (IPN) từ VNPay để xác nhận trạng thái thanh toán cuối cùng, đảm bảo tính idempotent khi cập nhật cơ sở dữ liệu.

SI-2: Dịch vụ bản đồ (Map Service API)

SI-2.1: Hệ thống EVCSMS phải tích hợp với Leaflet Map API để hiển thị vị trí các trạm sạc trên bản đồ.

SI-2.2: Hệ thống phải sử dụng Leaflet để hiển thị tọa độ (latitude, longitude) của từng trạm sạc và hỗ trợ người dùng tìm trạm sạc gần nhất.

SI-3: Giao tiếp nội bộ (Backend API)

SI-3.1: Giao diện web phải giao tiếp với API backend thông qua RESTful API (JSON over HTTPS) để truy xuất và cập nhật dữ liệu như người dùng, trạm sạc, và đặt chỗ.

SI-3.2: Các yêu cầu API phải có xác thực bằng JWT token để đảm bảo an toàn và phân quyền truy cập theo vai trò (Admin, Staff, User).

5.3 Hardware Interfaces

HI-1: Supported charger types include AC, Type 2, CCS, and CHAdeMO connectors with smart metering capabilities.

HI-2: The Admin Dashboard shall display hardware attributes such as connector ID, rated power (kW), and current status (e.g., *Available*, *Charging*, *Maintenance*) for monitoring and management.

5.4 Communications Interfaces

CI-1: All communications within the system are handled through standard HTTP requests between the web client and backend server.

CI-2: The system provides in-app notifications (via the notification bell icon) to inform users of reservation confirmations, charging status updates, and payment results.

6. Quality Attributes

6.1 Usability

USE-1: The system interface shall be intuitive enough for new users to navigate and perform key actions (e.g., view stations, make a reservation, start charging) without external guidance.

USE-2: All main user flows — such as station search, booking, and viewing charging history — shall be completed within a few steps and respond within 2 seconds for most actions.

USE-3: The map view shall display nearby charging stations accurately based on location and render results quickly for a smooth user experience.

USE-4: The system shall provide clear and readable notifications for both successful and failed operations, with retry options where applicable.

6.2 Performance

PER-1: User registration and login shall complete within ≤ 2 seconds under normal network conditions.

PER-2: Map rendering and initial station loading shall complete within ≤ 2 seconds for $\geq 90\%$ of requests.

PER-3: Booking requests shall be processed within ≤ 3 seconds.

PER-4: Booking or charging status notifications shall appear within ≤ 5 seconds after user action.

6.3 Security

SEC-1: The system shall use bcrypt with a cost factor of 12 to hash all user passwords before storing them in the database.

SEC-2: All users (Drivers and Admins) must authenticate through a secure login system before accessing restricted operations or personal data.

SEC-3: Admins may confirm payments made in cash and update the corresponding transaction status in the system.

SEC-4: Drivers shall be allowed to view and manage only their own transaction history, vehicle details, and charging sessions.

6.4 Safety

SAF-1: The system shall allow a charging session to start only after the user scans a valid QR code of the charging station.

SAF-2: The charging interface shall provide Pause, Resume, and Stop buttons so that users can safely control the charging process at any time.

SAF-3: When the user selects Stop, the system shall display a confirmation dialog to prevent accidental termination of the charging session.

6.5 Availability Requirements

AVL-1: The system shall operate stably under normal usage, except during server maintenance or development periods.

AVL-2: Temporary errors from the **payment simulation** or **map service (Leaflet)** shall not interrupt the entire system; users shall still be able to view and browse charging stations.

AVL-3: Important data such as **user accounts, booking history, and charging sessions** shall be securely stored in the database and restorable manually if needed.

6.6 Robustness Requirements

ROB-1: If the user loses connection, reloads the page, or closes the browser while performing actions such as booking or payment, the system shall temporarily save the current state and allow the user to recover or continue the operation when they return.

Appendix A:

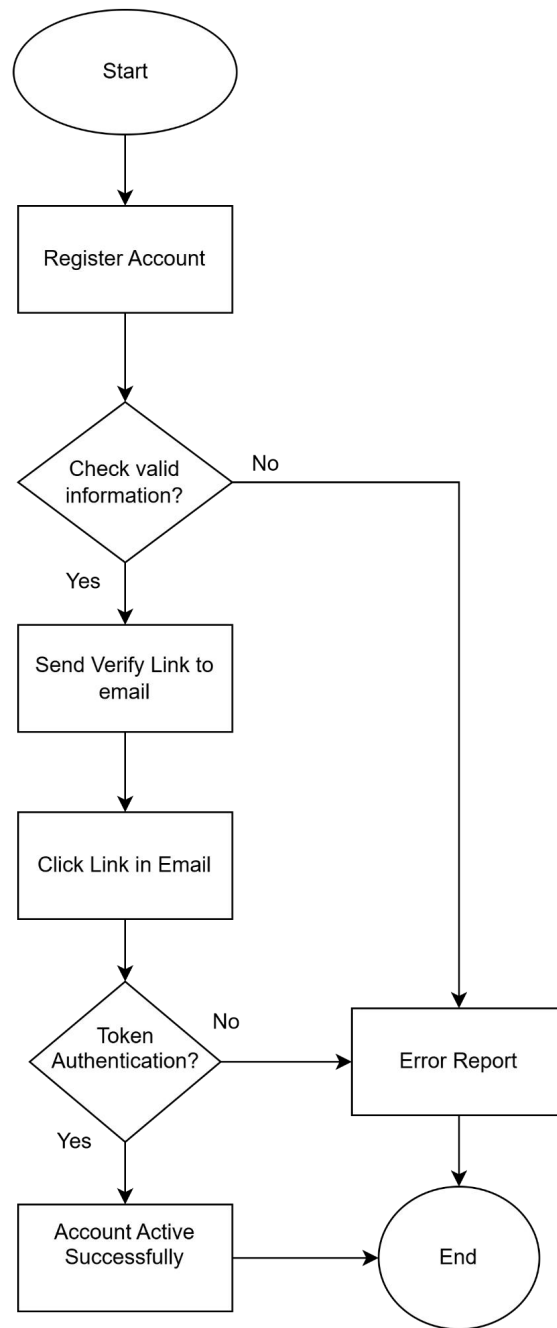


Figure 5. Main flow register for EV Charging Station Management System

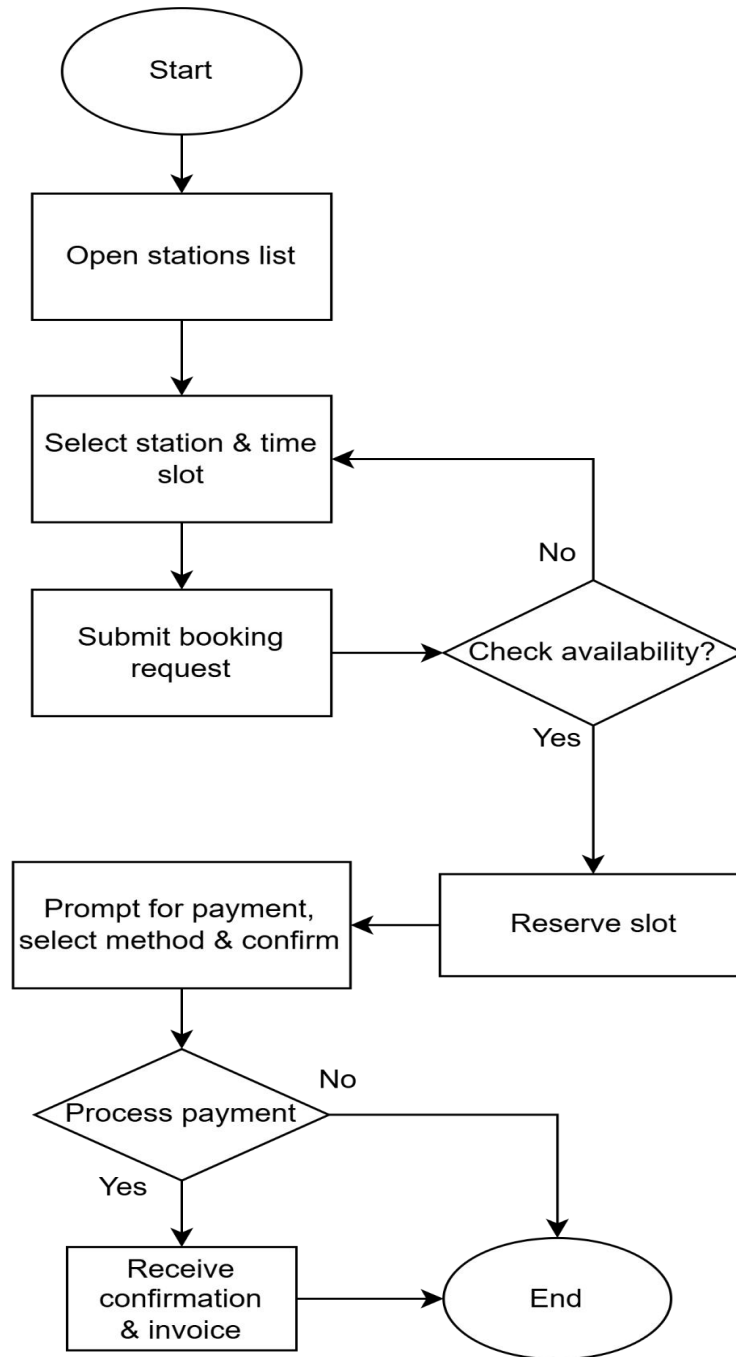


Figure 6. Main flow reservation for EV Charging Station Management System

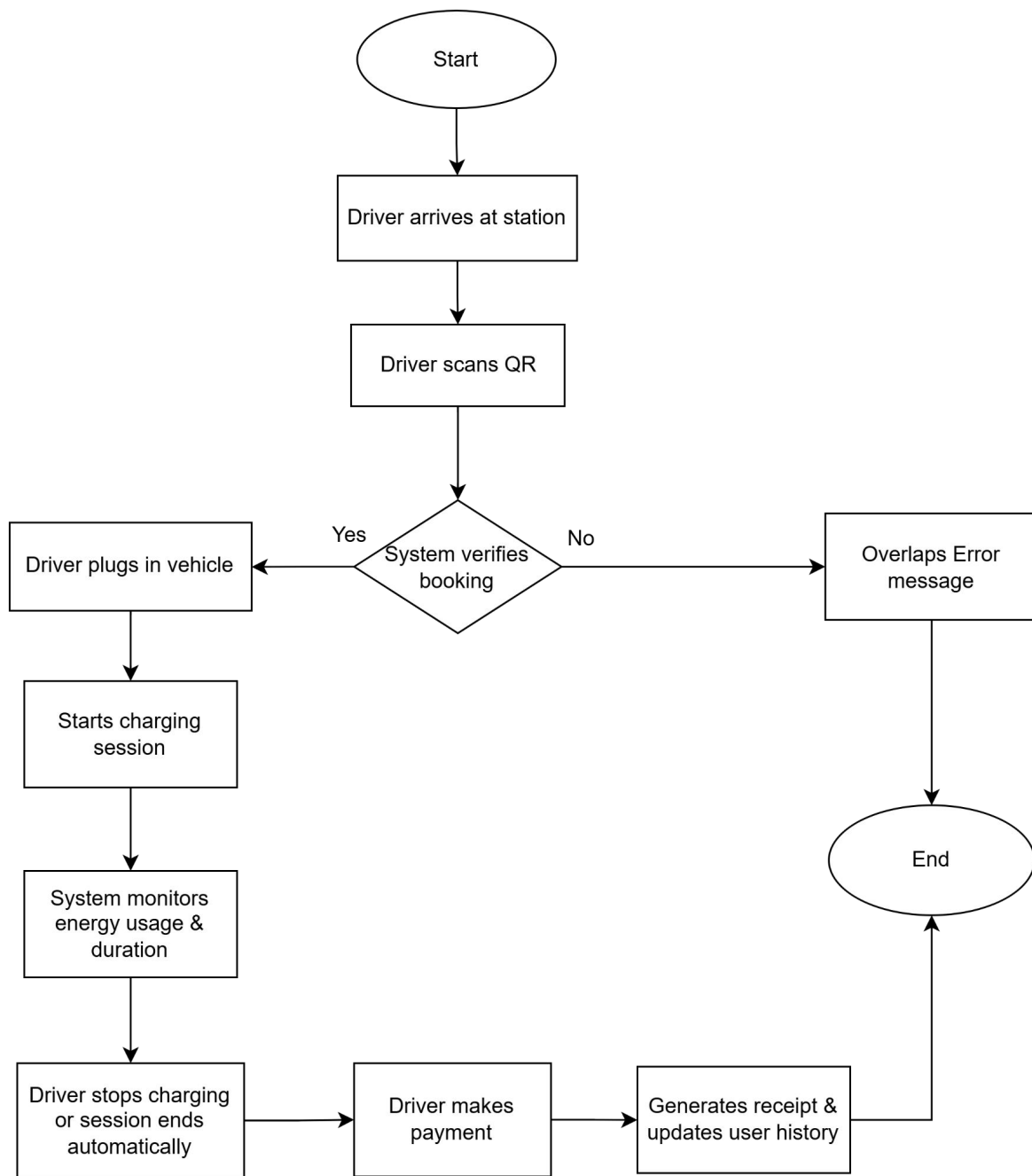


Figure 7. Main flow Charging Session for EV Charging Station Management System

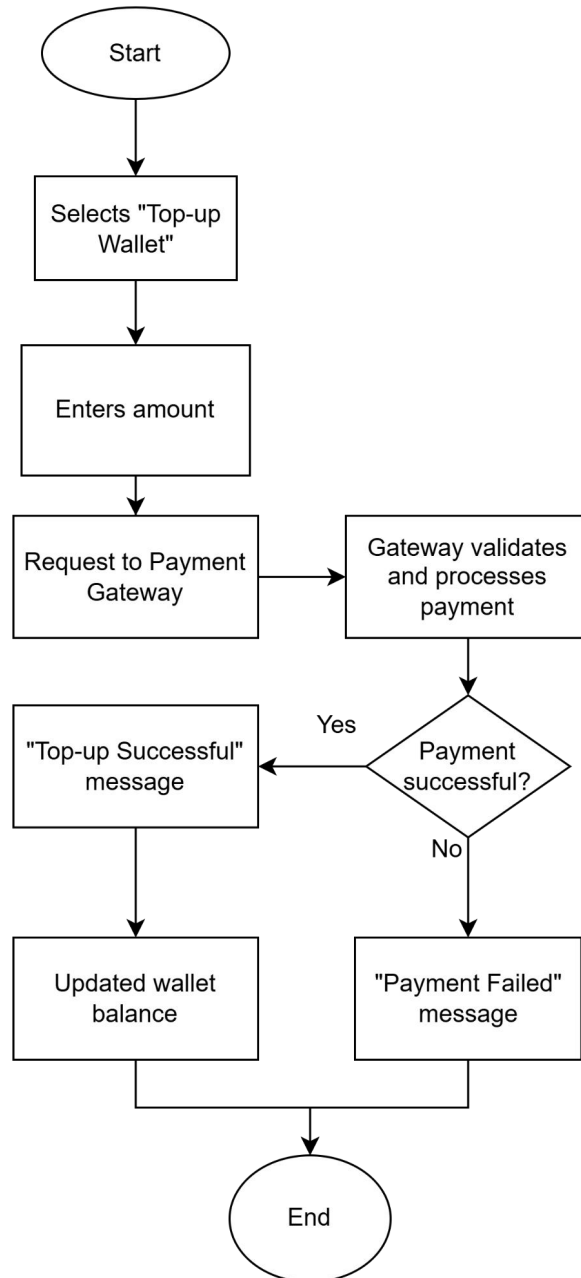


Figure 8. Main flow top up Prepaid payment for EV Charging Station Management System

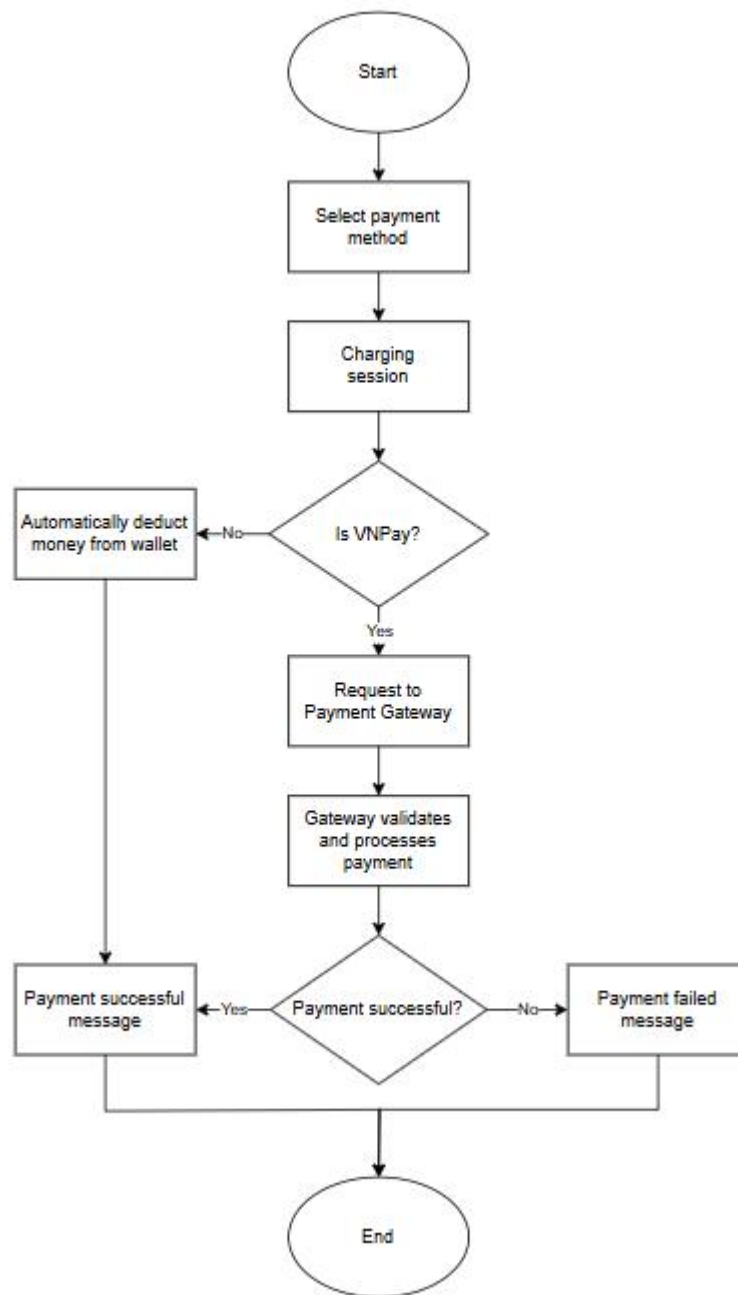


Figure 8. Main flow payment for EV Charging Station Management System

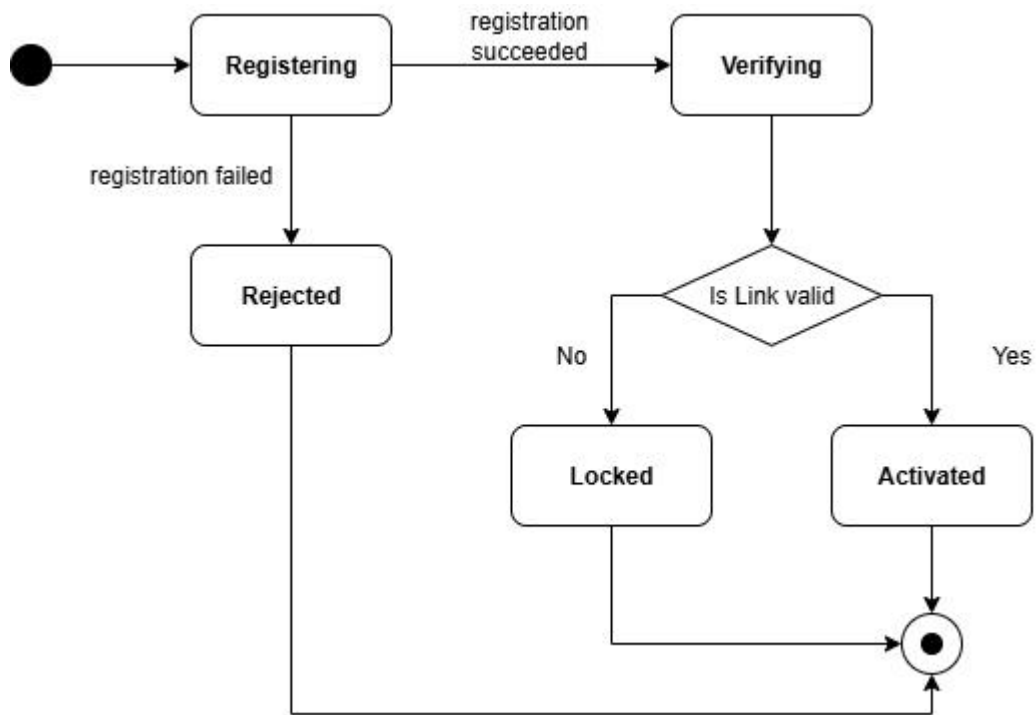


Figure 9. Register statechart for EV Charging Station Management System

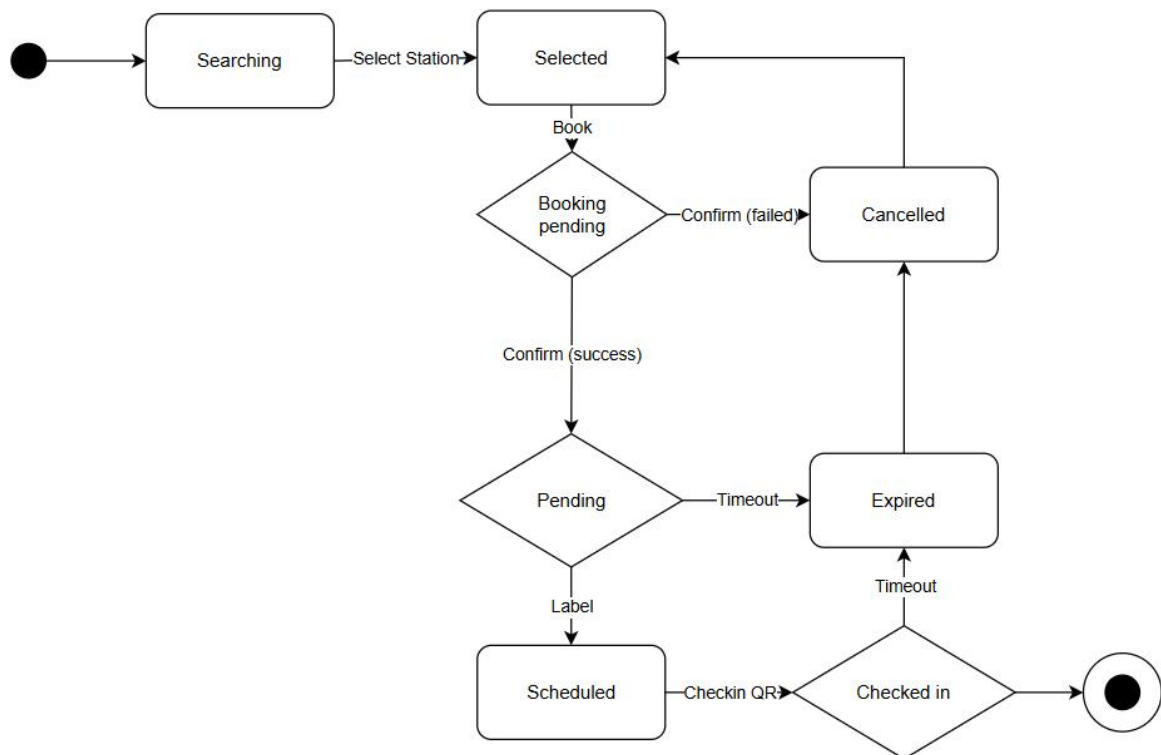


Figure 10. Reservation statechart for EV Charging Station Management System